M 155/78

Enrolment Number										

Total No. of printed pages = 03

Monsoon, 2024

B. Tech 3rd Semester Examination

Fluid Mechanics

Course Code: BCE23305T

Full Marks – 50 **Time – 2 hours**

The figure in the margin indicates full marks for the questions.

SECTION 1

Answer all the questions **Marks: 10 x 1 =10**

1. (a) The mass per unit volume of a liquid at standard temperature and pressure is called
   (i) specific weight (ii) mass density
   (iii) specific gravity (iv) none of the above

   (b) The values of coefficient of discharge, Cd for running full and running free mouthpiece are
   (i) 0.707 and 0.5 (ii) 0.5 and 0.4
   (iii) 0.4 and 0.707 (iv) 0.379 and 0.855

   (c) The surface tension in a soap bubble of 40 mm diameter when the inside pressure is 2.5 N/m^2 above the atmospheric pressure is
   (i) 1.25 N/m (ii) 0.125 N/m
   (iii) 0.0125 N/m (iv) 0.00125 N/m

   (d) A rectangular floating body is 20 m long and 5 m wide. The water line is 1.5 m above the bottom. If the centre of gravity is 1.8 m from the bottom then the metacentric height will be approximately
   (i) 3.3 m (ii) 1.65 m
   (iii) 0.34 m (iv) 0.30 m

   (e) Bernoulli's theorem deals with the principle of conservation of
   (i) energy (ii) momentum
   (iii) mass (iv) force

   (f) Froude's Number is square root of the ratio of
   (i) viscous force to gravity force
   (ii) inertia force to viscous force

(iii) gravity force to viscous force

(iv) inertia force to gravity force

(g) An open cylindrical tank 2 m in diameter and 2 m high contains water up to a depth of 1.5 m. Then the maximum angular velocity at which it can be rotated about its vertical axis without any spill of water is

(i) $\sqrt{(2g)}$ (ii) $\sqrt{(g)}$

(iii) 2g (iv) g

(h) In MLT system the dimensions of specific volume would be

(i) L^3 (ii) ML^3

(iii) ML^{-3} (iv) $M^{-1}L^3$

(i) 1 litre of crude oil weighs 9.6 N. Calculate its specific weight.

(i) 9600 N/m^3 (ii) 978.6 kg/m^3

(iii) 0.978 (iv) 978 N/m^3

(j) Pitot tube is a device used in the flowing fluid for measuring

(i) discharge (ii) pressure

(iii) velocity (iv) kinetic energy

SECTION 2

Answer any one question from this section **Marks: 10**

2 (a) Two plates are placed at a distance of 0.15 mm apart. The lower plate is fixed while the upper plate having surface area 1.0 m^2 is pulled at 0.3 m/s. Find the force and power required to maintain this speed, if the fluid separating them is having viscosity 1.5 poise. Find the kinematic viscosity if the specific gravity of the fluid is 1.9. (6+4=10)

(b) The velocity distribution over a plate is given by $u = (2/3)\,y - y^2$ in which u is the velocity in m/sec at a distance of y m above the plate. Determine the shear stress at y = 0, 0.1 and 0.2 m. Take viscosity equal to 6 poise. Find the distance in metres at which shear stress is zero. (6+4=10)

SECTION 3

Answer any one question from this section **Marks: 10**

3 (a) A vertical sluice gate is used to cover an opening in a dam. The opening is 2 m wide and 1.2 m high. On the upstream of the gate, the liquid of specific gravity

1.45, lies upto a height of 1.5 m above the top of the gate, whereas, on the downstream side the water is available upto a height touching the top of the gate. Find the resultant force acting on the gate and position of centre of pressure. Find also the force acting horizontally at the top of the gate which is capable of opening it. Assume that the gate is hinged at the bottom. (10)

(b) If for a 2 – D potential flow, the stream function, Ψ is given by –

$$\Psi = 2xy$$

Find the velocity at the point P (2, 3). Find also the value of velocity potential, Φ at the point. (10)

SECTION 4

Marks: 10

Answer any one question from this section

4 (a) A tank has two identical orifices on one of its vertical sides. The upper orifice is 3 m below the water surface and lower one is 5 m below the water surface. If the value of Cv for each orifice is 0.96, determine the point of intersection of the 2 jets. (10)

(b) Derive the expression of maximum discharge over a broad crested weir. (10)

SECTION 5

Marks: 10

Answer any one question from this section

5 (a) The pressure difference Δp in a pipe of diameter D and length l due to viscous flow depends on the velocity V, viscosity μ and density ρ. Using Buckingham's Π theorem, develop an expression for Δp. (10)

(b) A ship 300 m long moves in sea water, whose density is 1030 kg/m^3. A 1: 100 model of this ship is to be tested in a wind tunnel. The velocity of air in the wind tunnel around the model is 30 m/s and the resistance of the model is 60 N. Determine the velocity of ship in sea water and also the resistance of the ship in sea water. The density of air is given as 1.24 kg/m^3. Take the kinematic viscosity of sea water and air as 0.012 stokes and 0.018 stokes respectively. (10)